Ira A. Fulton Schools of Engineering
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CSE 579: Knowledge Representation and Reasoning

Automated Warehouse Scenario Project

Purpose

For your course project, you will work individually to solve a real-world problem using logic programming. This project was included in the ASP Challenge 2019 and presents opportunities to demonstrate your ability to contribute to industrial applications of knowledge representation and reasoning, specifically industry application of answer set programming (ASP). This course project is a variation of the ASP Challenge because you will use Clingo as the programming language to solve the challenge. Please refer to your course's Welcome and Start Here module for more information and resources related to Clingo.

This project is broken into four (4) parts that have separate deliverables throughout the course and will account for 30% of your grade. It will therefore qualify for inclusion in your MCS Portfolio (optional).

Objectives

Learners will be able to:

- Demonstrate that you can perform automated reasoning about actions using KR tools.
- Applied the different constraints on the problem
- Be familiar with different instances (Scenarios)

Technology Requirements

- CSE579_course-project-report-kit_AAAI
- Clingo Download Tutorial - Github
- Clingo Online
- Project Packages
    - The package consists of the following elements:
        - description.pdf: a problem description
        - simpleInstances/: a folder containing five instances

Project Description

This course project is an exploration of a dynamic automated planning scenario, akin to operations in Amazon warehouses. The project centers on an automated warehouse system where robots navigate a rectangular grid to transport shelves and fulfill orders at picking stations. The project description, milestones, and evaluation criteria outlined in this document ensure your preparedness and success in completing the project.

Throughout the course, utilize this document as a pivotal reference for project guidance. Additionally, a video walkthrough is available to further aid your understanding.

Degree students may use this course's projects for the Request for Faculty Review: MCS Project Portfolio . This is optional and only for degree students wanting to use this course's projects as part of their portfolio degree requirement/specialization requirements.

Project Prompt

Scenario: Your task in this project is to simulate a simplified version of automated planning similar to operations in an Amazon warehouse. This involves robots transporting products to picking stations to fulfill orders. The warehouse is represented as a rectangular grid, and the robots can move horizontally or vertically between adjacent cells. To fulfill given orders, robots have to carry shelves with the required products to matching picking stations. The robots are flat, can move underneath shelves and pick them up. However, a robot that carries a shelf does not fit under another shelf anymore, so that shelves may need to be moved out of the way first.

Challenge: The overall goal of the task is to fulfill all orders in as little time as possible, where time is counted in steps and each robot may (but does not have to) perform one action per time step. You will write hard constraints to find the plan to fulfill all orders. This is a dynamic world problem, meaning that the status of each object may change over time (e.g., the location of each robot may change over time).

Preparation: You will start to write hard constraints from the first milestone and will learn how to write more complex hard constraints in Module 4 and Module 5.

Details: View problem details and access example instances by downloading the project package.

Directions

Requirements and deliverables are detailed in the models they appear. At a glance, the project schedule is included in this overview. In general, part 1 and part 2 ensure that you are ready to run logic programs for this project, part 3 checks your midterm progress, and part 4 requires the final project report.

Each milestone has an overview document detailing the specific requirement for that milestone.

Submission Directions for Project Deliverables

Each part will have a submission page in the course. Please follow the directions related to each submission to submit your assignment. Some milestones may ask you to complete a series of multiple-choice questions and submit pdf files.

In addition to the downloaded package for the project, you will need to use a report kit as the template to write your project reports. In addition, you will be submitting files via and responding to prompts in staff-graded assignments. Please download the "Project Report Kit" that follows. It is adapted from the Author Kit used by the Association for the Advancement of Artificial Intelligence , the organization whose conventions and style manual you will be expected to use for the project reports.

Evaluation

In general, points will be awarded based on effort and the degree to which you adhere to the problem's requirements and follow its specifications regarding what has to be represented, computed, and delivered.

Detailed evaluation criteria are provided for each milestone as you progress through the project in the "Review Criteria" section (or an equivalent) for each deliverable, usually in the instructions for each submission or other milestone task. Below is a high-level summary of the evaluation criteria.

Milestone	Deliverable(s)	Points available	Evaluation Criteria

1	* Automated Warehouse Scenario Project Part 1	15	*Completion
2	* Automated Warehouse Scenario Project Part 2	45	*Completion *Standards met
3	* Automated Warehouse Scenario Project Part 3	90	*Completion *Standards met *Progress made
4	* Automated Warehouse Scenario Project Part 4	150	*Completion *Standards met